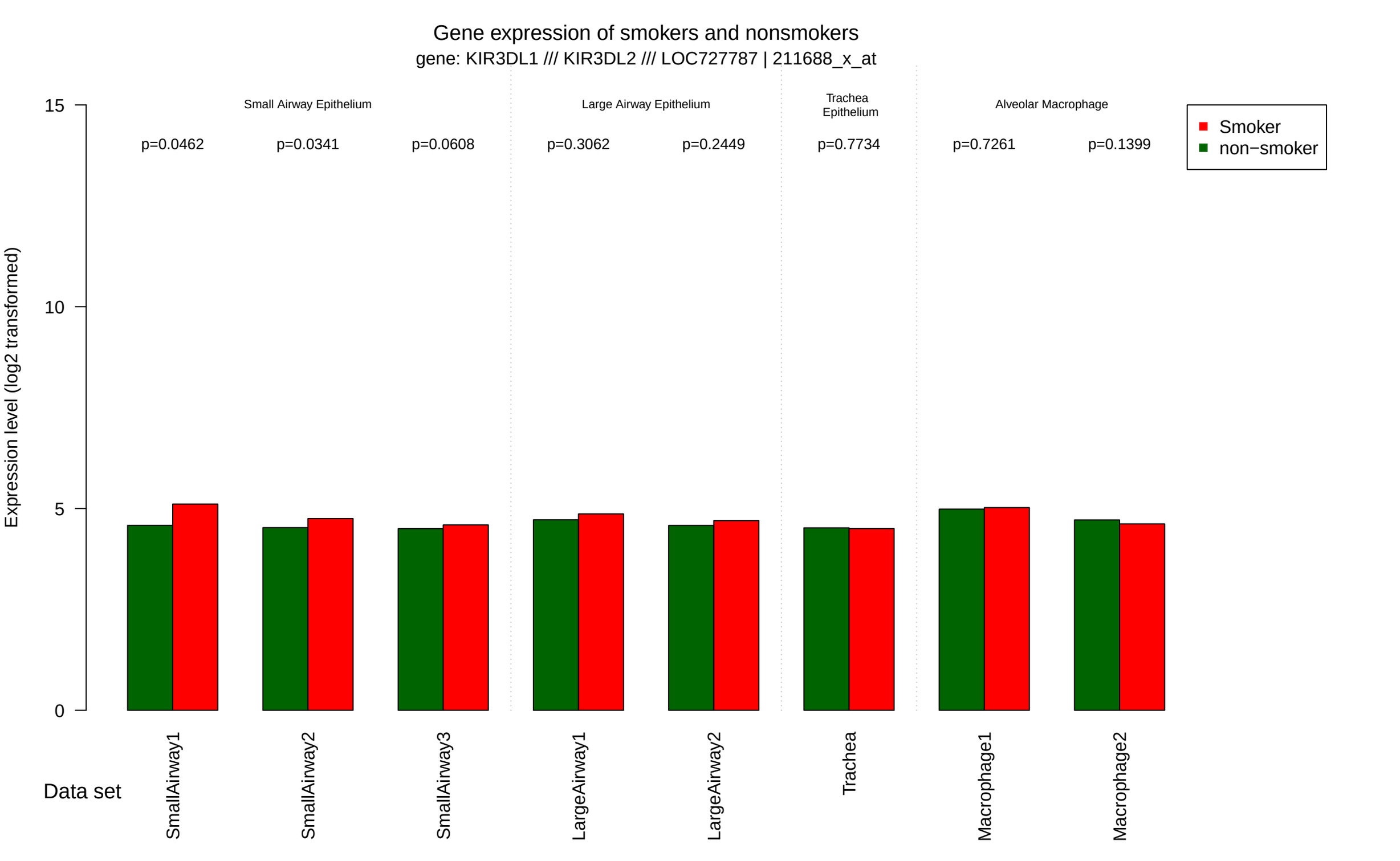
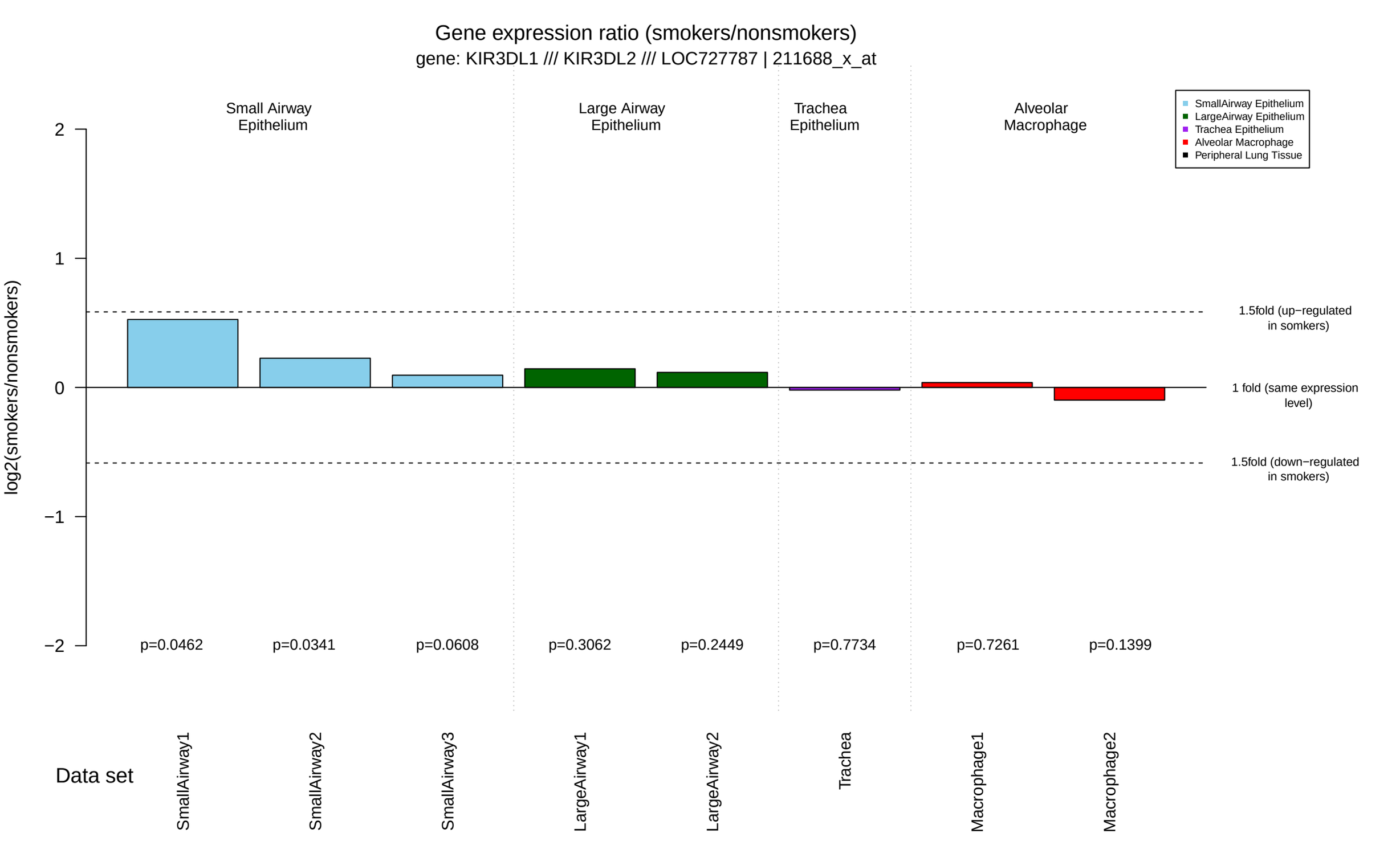
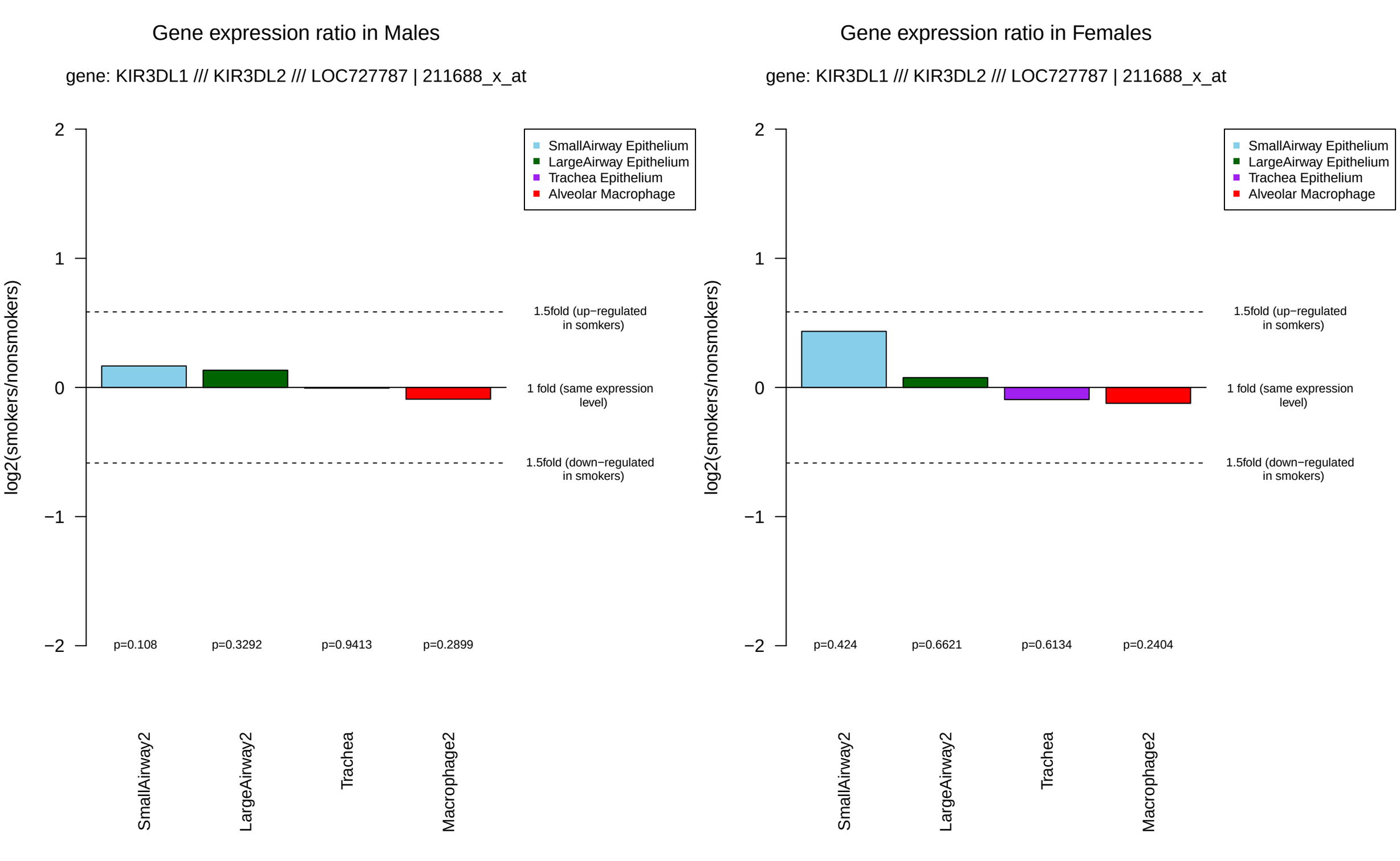
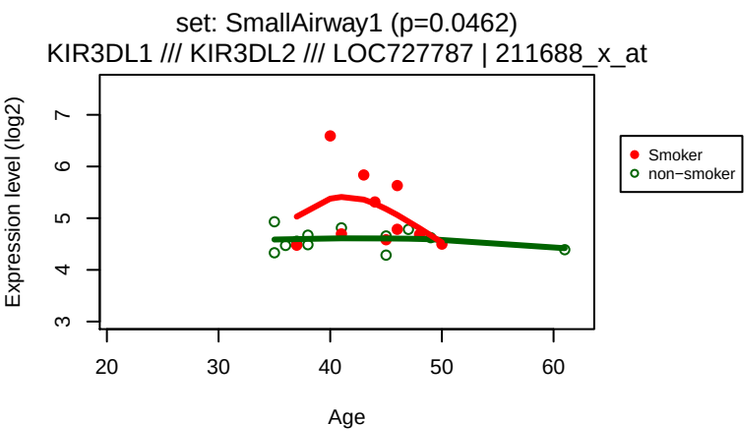
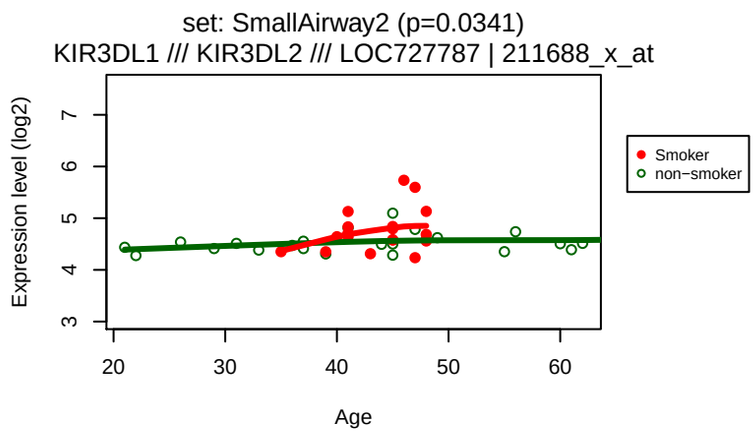
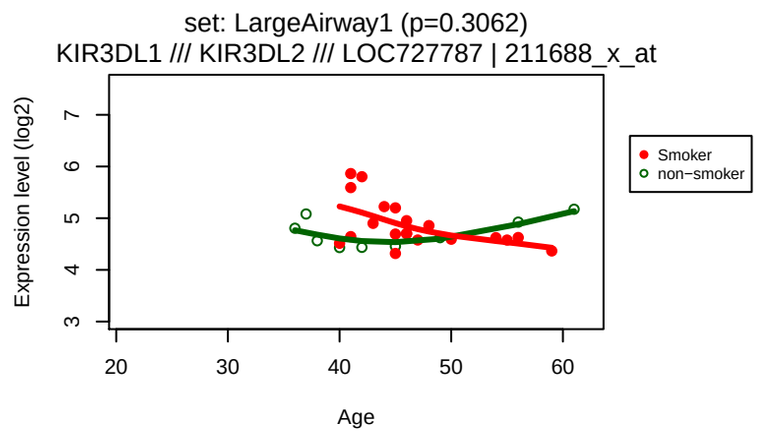

Gene expression of smokers and nonsmokers
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0462
p=0.0341
p=0.0608
p=0.3062
p=0.2449
p=0.7734
p=0.7261
p=0.1399
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

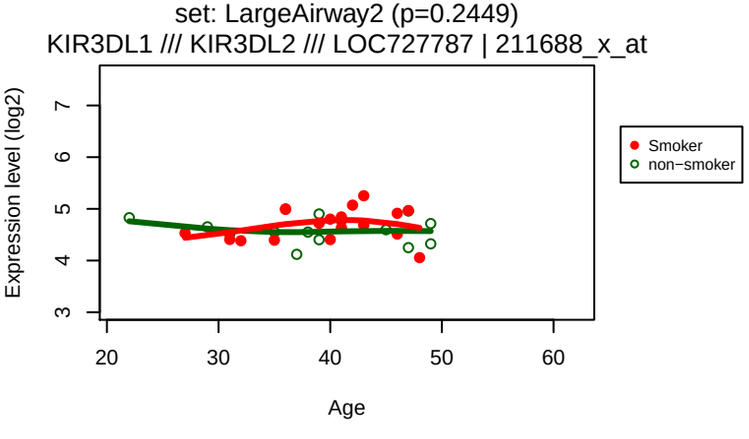

Gene expression ratio (smokers/nonsmokers)
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0462
p=0.0341
p=0.0608
p=0.3062
p=0.2449
p=0.7734
p=0.7261
p=0.1399
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

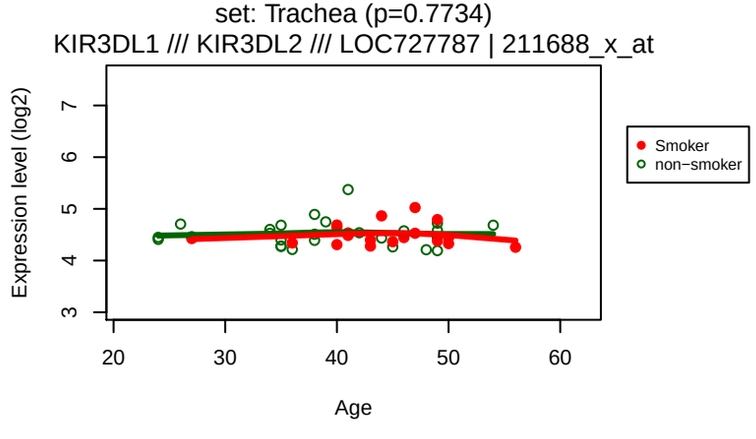

Gene expression ratio in Males
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.108
p=0.3292
p=0.9413
p=0.2899
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.424
p=0.6621
p=0.6134
p=0.2404
SmallAirway2
LargeAirway2
Trachea
Macrophage2

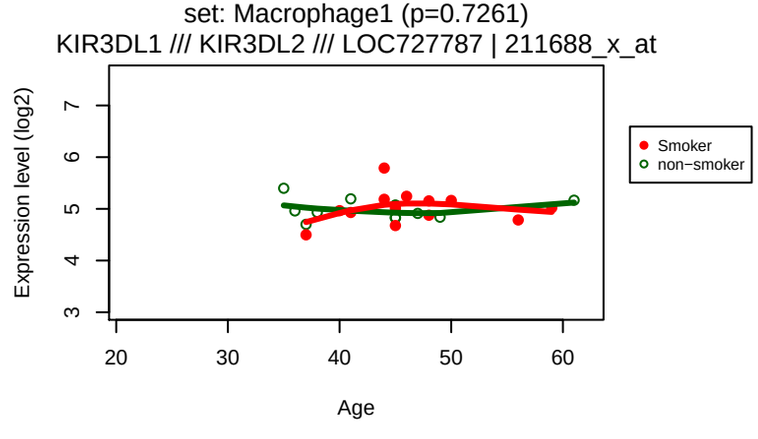

set: SmallAirway1 (p=0.0462)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Expression level (log2)
Age
Smoker
non−smoker

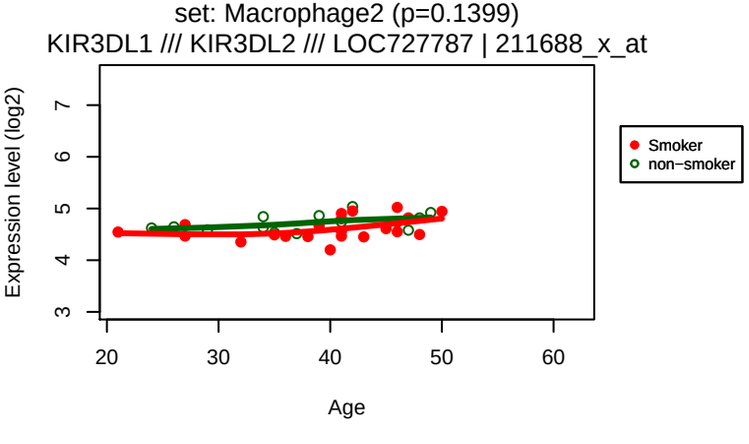

set: SmallAirway2 (p=0.0341)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Expression level (log2)
Age
Smoker
non−smoker

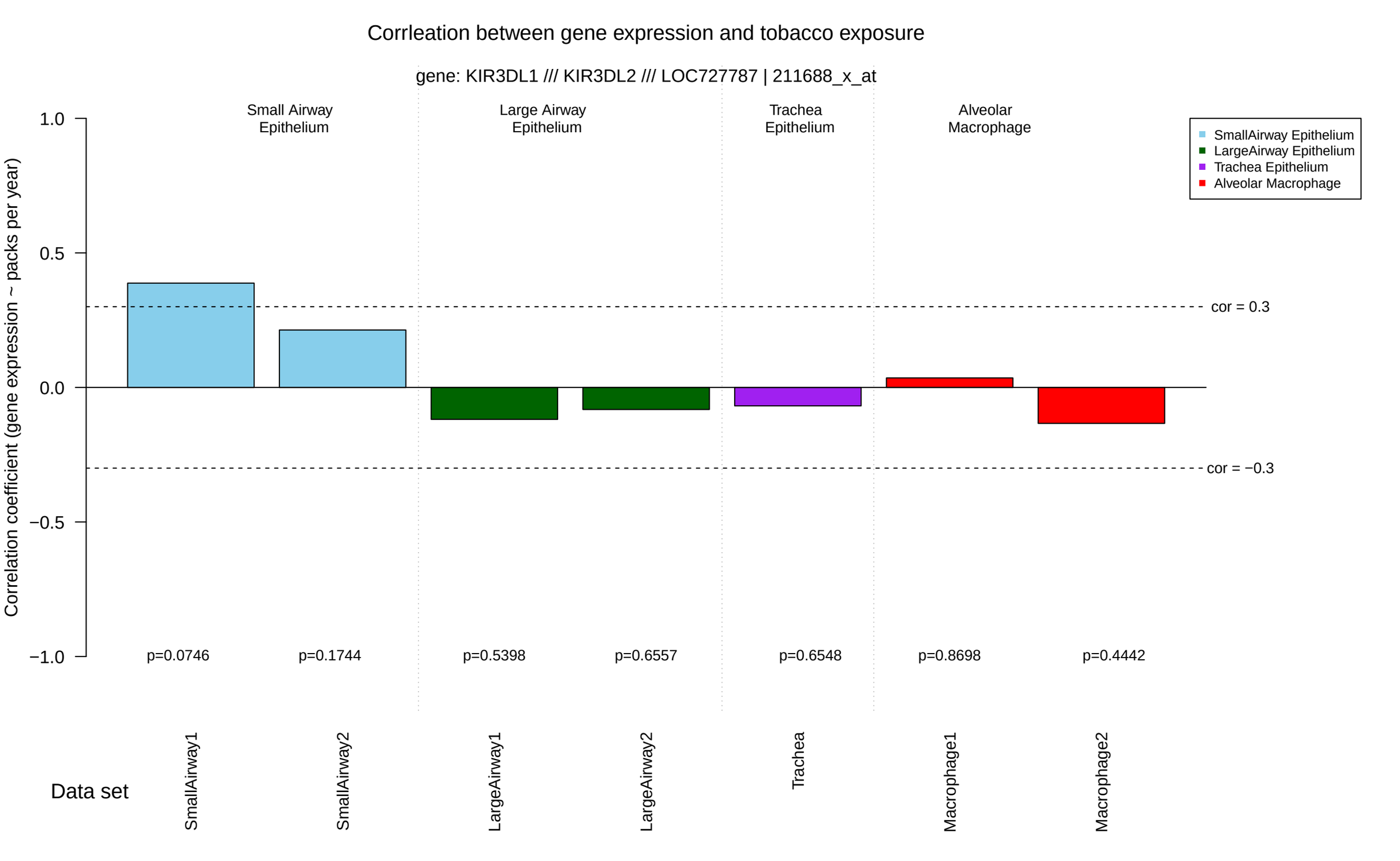

set: LargeAirway1 (p=0.3062)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2449)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7734)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7261)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1399)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 211688_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0746
p=0.1744
p=0.5398
p=0.6557
p=0.6548
p=0.8698
p=0.4442
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2